

FRANCESCO SARNO

Senior Computer Vision Engineer
Applied AI Researcher

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Professional Summary

Senior Computer Vision Engineer and Applied AI Researcher with 5+ years of experience in 3D reconstruction, pose estimation, robotics, and medical imaging. Led R&D from prototype to deployment, built production computer vision systems, and secured competitive research funding for applied projects in industrial and biomedical environments.

Core Competencies and Technical Skills

- **Computer Vision & 3D Perception:** 3D Reconstruction, Multi-View Geometry, Structure from Motion (SfM), Simultaneous Localization and Mapping (SLAM), Pose Estimation, Camera Calibration, Photometric Stereo, Neural Radiance Fields (NeRF), Detection, Segmentation, Tracking, Optical Character Recognition (OCR), Real-Time Vision Systems.
- **Machine Learning & AI:** Deep Learning, Generative AI, Diffusion Models, Generative Adversarial Networks (GANs), Vision-Language Models (VLMs), Large Language Models (LLMs), Graph Convolutional Neural Networks (GCNNs), Deep Reinforcement Learning, Neural Architecture Search (AutoML), Synthetic Data Generation.
- **Robotics & Simulation:** ROS2, Robot Perception, Sensor Fusion, Multi-Camera Systems, Robot-Mounted Vision Systems, NVIDIA Omniverse, Simulation-Based Training, Digital Twins.
- **Programming & Software Engineering:** Python, PyTorch, OpenCV, Open3D, C++, C, C#, MATLAB, Unity, Linux, Docker, Git, Bash, Conda, UV, AWS, Microsoft Azure.
- **Research & Technical Leadership:** Applied R&D, Experimental Design, Scientific Publishing, Technical Project Leadership, Proposal Writing (European and Innosuisse Projects).
- **Languages:** Italian (Native), English (C2), Spanish (B1), German (A2), French (A1).

Professional Experience

X23D AG Zurich, Switzerland
Senior Computer Vision and AI R&D Engineer Jul 2025 - Present

- Lead computer vision R&D for 3D human-bone reconstruction from X-ray imagery, improving the core medical imaging pipeline for deployment-ready use in a production product.
- Designed and developed deep learning pipelines for accurate anatomical reconstruction from sparse X-ray data, increasing reconstruction quality and reducing manual intervention.
- Built a synthetic data generation framework to mitigate data scarcity and improve model robustness across edge cases and unseen anatomical variations.
- Designed high-precision camera and imaging calibration pipelines for complex X-ray acquisition systems.

Switzerland Innovation Park Biel/Bienne - Swiss Battery Technology Center Biel, Switzerland
Computer Vision and Robotics Lead Research Engineer Feb 2023 - Aug 2025

- Led the development and submission of European Commission and Innosuisse proposals, helping secure funding for multiple R&D initiatives and expanding the lab's applied research scope.
- Designed production-ready detection, segmentation, 3D reconstruction, and pose estimation systems for EV battery components using multi-view geometry and deep learning, enabling automated inspection workflows.
- Integrated real-time computer vision models within ROS2 workflows to support autonomous manipulation.
- Developed synthetic data generation pipelines in NVIDIA Omniverse to accelerate model development and validation for robotic perception and inspection tasks.
- Applied deep reinforcement learning in simulation environments for autonomous robotic task execution.
- Configured and calibrated fixed and robot-mounted multi-camera systems for 3D capture and synchronization.

- ETH Juniors**
Computer Vision Part-Time Consultant

Zurich, Switzerland
Sep 2023 - Present

 - Lead multiple computer vision proof-of-concept projects, managing the complete ML lifecycle from prototype to deployment.
 - Developed defect detection and localization systems for challenging industrial surfaces using OpenCV, PyTorch, and custom CNN architectures, improving inspection reliability in difficult imaging conditions.
 - Generated synthetic 3D environments using VLMs and simulation platforms to support model training and validation.
 - Designed OCR and information extraction pipelines for automated fraud detection and document processing.
- EPFL - Computer Vision Laboratory**
Computer Vision Research Engineer

Lausanne, Switzerland
Feb 2022 - Mar 2023

 - Conducted research in biomedical imaging, 3D reconstruction, and GCNNs for human organ reconstruction.
 - Developed algorithms for SfM, SLAM, and camera calibration in research-oriented computer vision systems.
- ETH Zürich - Computer Vision Laboratory**
Computer Vision Research Assistant

Zurich, Switzerland
May 2021 - Dec 2021

 - Conducted research in NeRF, deep photometric stereo, and automated machine learning.
 - Co-authored two peer-reviewed papers accepted at WACV 2022, demonstrating publication output in a competitive CV venue.
- Solera Holdings (Qapter) via ETH Juniors**
Computer Vision Engineer

Zurich, Switzerland
Aug 2021 - Dec 2021

 - Developed deep learning and NeRF-based pipelines for 3D vehicle surface reconstruction.
 - Implemented semantic segmentation and damage quantification systems for automotive inspection workflows.
- Rheinmetall Air Defence**
Computer Vision Engineer Intern

Zurich, Switzerland
Feb 2020 - Dec 2020

 - Developed computer vision algorithms for firing accuracy assessment and ballistic trajectory visualization.
 - Created interactive 3D visualization tools for performance analysis.

Education

- M.Sc. in Robotics, Systems and Control**
ETH Zürich
Advisor: Prof. Dr. Roland Siegwart
Grade: 5.61/6.00

2018 - 2021
- B.Sc. in Automation Engineering**
Politecnico di Milano
Grade: 106/110

2015 - 2018
- High School Diploma**
Liceo Scientifico Statale N. Copernico
Grade: 95/100

2010 - 2015

Publications

- WACV 2022** - Neural Architecture Search for Efficient Uncalibrated Deep Photometric Stereo.
Francesco Sarno, Suryansh Kumar, Berk Kaya, Zhiwu Huang, Vittorio Ferrari, Luc Van Gool.
- WACV 2022** - Neural Radiance Fields Approach to Deep Multi-View Photometric Stereo.
Berk Kaya, Suryansh Kumar, Francesco Sarno, Vittorio Ferrari, Luc Van Gool.

Certificates and Awards

- Best Poster Presentation Award, NVIDIA GTC 2025.
- ICVSS 2024: Computer Vision in the Age of Large Language Models.
- IEEE RAS Summer School on Multi-Robot Systems, CTU Prague, 2022.
- IELTS Academic, Overall Score: 7.5 (2018).

Selected Projects

Master's Thesis

ETH Zürich, Computer Vision Lab

Advisor: Prof. Dr. Luc Van Gool, Dr. Suryansh Kumar, Dr. Berk Kaya

Completed with distinction 5.75/6.00

Exploring Automated Machine Learning Framework for Deep Photometric Stereo: developed an automatically designed pipeline that achieves state-of-the-art results in uncalibrated photometric stereo.

Semester Project

ETH Zürich, Autonomous Systems Lab

Advisor: Prof. Dr. Roland Siegwart, Dr. Abel Gawel, Dr. Hermann Blum

Semantically informed localization in building structures: developed a pipeline for localization of a four-wheel robot in indoor environments using a segmentation-oriented neural network and point clouds.

Course Project

ETH Zürich, Computer Vision and Geometry Group

Advisor: Prof. Dr. Marc Pollefeys

Fully Convolutional Place Recognition Network: developed an algorithm for sparse SLAM with point clouds in large outdoor environments.

Course Project

ETH Zürich, Innovation Center Virtual Reality

Advisor: Prof. Dr. Andreas Kunz

Virtual Reality Game Development Project: created a maze-based video game from scratch, playable with keyboard and HTC VIVE.

Volunteering

- Football Coach, FC Italia Zurich (2025 - Present).
- Volunteer, Gruppo Volontariato Primavera (2022 - Present).
- Football Coach, GSO Azzano Mella (2022 - 2024).
- Core Team Member, Google Developer Student Club Zürich (2021 - 2022).

References

Prof. Dr. Luc Van Gool, Prof. Dr. Pascal Fua, Prof. Dr. Roland Siegwart, Prof. Dr. Suryansh Kumar, Dr. Berk Kaya, Dr. Udaranga Wickramasinghe.